BioRaman

Raman Hyperspectral Map Analysis for Biophysics — User Manual

Version 0.10.0 · application file: **bioraman.py**

Desktop GUI (Python / Tkinter) · detailed step-by-step edition

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A desktop application to load, visualise and analyse Raman hyperspectral maps and confocal volumes: multi-format import, reproducible preprocessing recipes, batch & headless processing, univariate / ratio mapping, curve fitting, PCA, clustering, MCR-ALS, N-FINDR, peak identification, quality control, and a publication-quality 3-D volume renderer with reference-spectra concentration analysis.

★ This edition adds **numbered step-by-step instructions for every tab** (Part II) and a greatly expanded **3-D Volume & Concentration** chapter (Part III) — see sections 19–21.

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Appendix B — metadata columns

Part I — Getting started

1 Introduction

BioRaman works with Raman hyperspectral maps — a full spectrum at every pixel of a 2-D area, or a 3-D confocal volume. The main window shows the map on the left and the spectrum at the cursor on the right; every analysis opens in its own window from the toolbar or menu, so several can stay open at once.

★ New in v0.10.0: reproducible recipes, non-destructive reprocess, processed-data export (NPZ/HDF5/CSV/MAT), batch & headless CLI, QC maps, HTML reports, sessions, true confocal-volume reconstruction, and a publication-quality 3-D renderer with concentration (CLS) analysis.

2 Installation & launching

Requires Python 3.9+. Install dependencies:

```
pip install numpy scipy matplotlib pybaselines renishawWiRE pillow

pip install scikit-learn pandas seaborn openpyxl
```

Optional (auto-installed on first use where possible): **ramanspy** (WITec .wip), **h5py** (HDF5 export), **plotly** + **kaleido** (3-D renderer + PNG export). Launch with:

```
python bioraman.py
```

Note On first launch BioRaman auto-installs the native **renishawWiRE** reader if missing — this is what recovers the true X×Y×Z geometry of .wdf maps and volumes. Set BIORAMAN_NO_AUTOINSTALL=1 to disable.

3 The main window & toolbar

The toolbar (two rows at the top) is the entry point to every feature. The buttons, left to right:

Button	Opens / does
Open file	Load any Raman map / volume / spectrum (§7)
Preprocess	Preprocessing settings & recipes (§8)
White Light	Load an optical reference image (§9)
PCA	Principal Component Analysis (§15)
3D Volume	3-D confocal viewer (§19)
Univariate	Single-band intensity / area maps (§10)
Dynamic / Ratio	Live band & band-ratio maps (§11)
Curve Fit	Per-pixel peak-fit maps (§12)
LUT / Profiles	Colour control; line profiles (§13)
ROI Analysis	Region-of-interest workflow (§14)
Cluster / MCR-ALS / N-FINDR	Unsupervised unmixing (§16–17)
Spectral Tools	Resample / crop / rotate / subtract (§18)
✓ QC Maps	Per-pixel quality control (§18)
■ HQ Volume	Publication 3-D renderer + concentration (§20–21)
■ Batch	Process a whole folder (§18)
Save Map / Spec / Processed	Export (§23)

Top-right: **MODE** (Ratio / A+B RGB / White Light) and **CMAP** (colour map) selectors, plus a progress bar.

4 Loading data & file formats

Routed automatically by extension. .wdf maps/volumes are read natively by renishawWiRE, which recovers the real grid (and depth axis for volumes).

Format	Backend	Notes
.wdf	renishawWiRE	Renishaw maps, line scans, single spectra, and confocal volumes (X·Y·Z).
.wip	ramanspy	WITec; pip install ramanspy.
.txt .csv .dpt .dat .jdx .asc	built-in	ASCII; auto delimiter; Renishaw #X #Y #Wave #Intensity long format.
.xlsx .xls	pandas	Tabular spectra (PCA input).
.zip	built-in	RAMANMETRIX dataset + metadata (§5).

Note If a map loads as a thin strip (e.g. *15625×1*), renishawWiRE is missing — install it and reload; the status bar should then show the real grid (e.g. *125×125* or *25×25*).

5 RAMANMETRIX datasets

Open **File** → **RAMANMETRIX Dataset (ZIP + metadata)** to import a ZIP of spectra plus a metadata table (filename containing "metadata"). Generate a template, apply labels (longest-pattern-wins, include filtering, * and . wildcards), or load any single spectrum into the viewer.

6 Preprocessing & recipes

Preprocessing runs automatically on load, in this order: cosmic-ray removal → dark subtraction → baseline → Savitzky–Golay smoothing → normalisation.

★ For quantitative concentration analysis (§21) set **Normalisation = none** so the map and your reference spectra share one intensity scale.

Recipes & reprocess

1. Open **Preprocess** and set the options; click **Apply & Close**.
2. Save them with **Preprocessing** → **Save Recipe…** (a .json file).
3. Re-use later with **Load Recipe…**; tick to re-apply immediately.
4. After changing settings, click **Preprocessing** → **Reprocess** to re-run on the retained raw data — no need to reload the file.

Part II — Step-by-step for every tab

Each section lists the exact clicks. Load a file first (§7) unless noted.

7 Open file

1. Click **Open file** (or Ctrl+O).
2. Choose your .wdf / .wip / .txt / .csv / .zip file.
3. Wait while it loads and preprocesses (progress bar fills).
4. Read the status bar: map size, wavenumber range, cosmic rays removed, time.
5. Click any pixel on the map to show its spectrum on the right; right-click to add a spectrum to the comparison overlay.

8 Preprocess

1. Click **Preprocess**.
2. Set **Cosmic-ray** removal (threshold/width) to clean spikes.
3. Choose a **Baseline** method (asls / arpls / drpls / none) to remove fluorescence background.
4. Enable **Smoothing** (Savitzky–Golay window & polynomial order).
5. Pick **Normalisation** (max / area / SNV / vector / none).
6. Click **Apply & Close**, then **Reprocess** to apply to the loaded data. Open **Processing Log** to confirm what ran.

9 White Light

1. Click **White Light** and select an optical image (or it loads automatically from a .wdf when embedded).
2. Switch **MODE** → **White Light** to view it, or use the LUT window (§13) to blend the chemical map over it with the opacity slider.

10 Univariate

1. Click **Univariate**.
2. Choose **Map type**: peak intensity, integrated band area, or single-point intensity.
3. Set the band limits in cm^{-1} with the sliders/boxes.
4. Click **Create Map**; the main map updates. Adjust CMAP and LUT as needed.
5. Save via **Save Map** (§23).

11 Ratio & Dynamic

1. Set **MODE** → **Ratio** (single band) or use the **Ratio** tab for a band-ratio A/B map.
2. Define **Band A** and **Band B** ranges (cm^{-1}).
3. Use **Dynamic** to drag the band live and watch the map update.
4. Set **MODE** → **A+B RGB** to render two bands as colour channels.

12 Curve Fit

1. Click **Curve Fit**.
2. Set minimum peak **width** and **height** to avoid spurious fits.
3. Choose the parameter to map (position / width / amplitude).
4. Click **Run Fit & Create Map** (slow on large maps — every pixel is fitted).

13 LUT & Profiles

1. Click **LUT** to open colour control.
2. Pick a colour scheme; set **vmin/vmax** manually or keep auto; set **opacity** to blend over the white-light image.
3. Click **Profiles**, then draw a line on the map to plot an intensity profile along it; export the profile as needed.

14 ROI Analysis

1. On the main map click **Draw ROI**, choose a shape (rectangle / ellipse / polygon / freehand) and draw. Tick **Mask outside ROI** to isolate it.
2. Click **Analyse ROI**.
3. Define the characteristic **bands** (A, B, …).
4. Click **Auto-set threshold** (or set manually) to separate signal from background; enable rubber-band baseline for weak bands.
5. Click **Run Analysis** for binarized maps, heatmaps and per-region stats.
6. Use **Export Figure / Export Panels** to save.

15 PCA

1. Click **PCA**.
2. Set **Components** (2–10) and the wavenumber window.
3. Optionally sub-sample spectra/file for speed, remove >2σ outliers, and standardise features.
4. Click **Run PCA**. Read the five panels: scores, loading spectrum, score distribution, scree (explained variance), and the spatial score map.
5. Use the **Display PC** spinner to switch components; **Save PCA Figure** to export.

16 Cluster

1. Click **Cluster**.
2. Choose **K-means** or **Agglomerative** and the number of clusters (2–10).
3. Click **Run**. A colour-coded class map and per-cluster mean spectra appear.
4. Read the **silhouette score** in the status line (–1→1; higher = better separation) to judge the cluster count.
5. Export the cluster map (PNG) or label matrix (CSV).

17 MCR-ALS & N-FINDR

MCR-ALS (unsupervised unmixing)

1. Click **MCR-ALS**; set the number of components and non-negativity constraints; optionally bin pixels for speed.
2. Click **Run MCR-ALS** for abundance maps + recovered pure spectra.
3. Use **Compare Components / Identify Peaks / Save Pure Spectra (CSV)**.

N-FINDR (endmember extraction)

1. Click **N-FINDR**; set the number of endmembers and the wavenumber window.
2. Click **Run N-FINDR**; abundance maps are computed by NNLS.
3. Export endmembers (CSV) / figure.

> **Note** MCR-ALS and N-FINDR discover components from the data. To quantify against your own measured references, use Component Analysis (§17b).

17b Component Analysis (DCLS/NNLS) & concentration

Supervised analysis: fit each pixel as a least-squares sum of your **reference spectra** to get concentration maps, a lack-of-fit map and overall concentration estimates — the Renishaw WiRE "Component analysis" workflow.

> ★ For quantitative results load the map with **Normalisation = none** and choose **None (quantitative)** here; references should be measured under identical conditions.

1. Click **Analysis** → **Component Analysis (DCLS/NNLS)**.
2. Click ■ **Load reference spectra…** and select all components (any format).
3. Choose **Method**: NNLS (non-negative, recommended) or DCLS (fast, allows negative).

4. Choose **Spectrum / 1st / 2nd derivative** (derivatives remove background; use for high-quality data).
5. Choose **Normalise** = None for quantitative, or Vector / Mean-centre for qualitative.
6. Set the **Background** polynomial order (Spectrum method) and tick **Calculate % lack of fit** to find unaccounted components.
7. Click **Run**. Each component concentration map and the LoF map are shown; the **concentration estimate** (% per component) appears on the left.
8. Use **Particle statistics on component** to quantify domains, or **Export maps + estimates** to CSV.

> **Note** Lack-of-fit (LoF) highlights pixels poorly explained by your references — inspect bright LoF regions, save those spectra as new references, and re-run until LoF is low (the WiRE 'find unknowns' workflow).

17c Particle Statistics

Quantify particles / domains in any single-band or concentration map (area, equivalent circle diameter, counts).

1. Build a map (Univariate / Ratio / a Component Analysis concentration map).
2. Open **Analysis** → **Particle Statistics** (or the button in Component Analysis).
3. Keep **Auto-binarise (Otsu)**, or untick it and set the threshold slider.
4. Tick **Remove edge particles** and set a **Min size** to filter noise.
5. Set the **Pixel size (µm)** so diameters are in microns.
6. Read the summary (count, area %, mean / median ECD) and the size histogram; **Export particle table (CSV)** for per-domain values.

18 Spectral Tools · QC Maps · Save · Batch

Spectral Tools

- **Resample** to an equidistant grid · **Crop** to a pixel box · **Rotate** 0/90/180/270° · **Background subtraction**. Applied in place — use Reprocess or reload to undo.

QC Maps

1. Click ✓ **QC Maps**.
2. Pick a metric: SNR, total intensity, max intensity, or raw-max saturation.
3. Read the min/median/mean/max summary; **Save QC map** to CSV/NPY.

Save Map / Spectrum / Processed

- **Save Map** (Ctrl+M) → PNG / TXT / NPY.
- **Save Spectrum** (Ctrl+S) → TXT / CSV of the selected pixel.
- **Save Processed** (Ctrl+Shift+S) → the full preprocessed cube as NPZ / HDF5 / CSV / TXT / MAT (re-loadable).

Batch

1. Click ■ **Batch**.
2. Choose input and output folders and the output format.
3. Optionally recurse into sub-folders and load a recipe.

4. Click **Run batch**; each file is processed and a batch_summary.csv written.

18b Library Search (identify a spectrum)

Analysis → **Library Search** identifies an unknown spectrum by correlating it against a reference library you supply. **No database is bundled** with BioRaman — download a free one and point the tool at it.

> ★ Free libraries: **RRUFF** (minerals, rruff.info), **Raman Open Database** (CC0), and **SLoPP / SLoPP-E** (microplastics). Observe each library's licence and cite it.

1. Click **Analysis** → **Library Search**.
2. Click ■ **Load library folder…** and choose the folder of reference spectra (or ■ **Load library files…**). The count of loaded spectra appears.
3. Choose the **Query spectrum**: Selected pixel, ROI mean, or Whole-map mean.
4. Pick **Preprocess** = SNV or 1st derivative (reduces baseline / intensity bias; recommended) and the number of **Top matches**.
5. Click ■ **Search**. The ranked table shows each match name and its Pearson correlation (r); click a row to overlay query vs match.
6. Click ↓ **Export results (CSV)** to save the ranked hits.

> **Note** Matching is computed only over the wavenumber range the library and your data share, so fingerprint-only libraries work even with CH/OH-range data.

Part III — 3-D Volume & Concentration ★

★ This is the highlighted part of the manual. BioRaman reconstructs a true confocal volume from depth-resolved .wdf files and renders it to publication quality, including quantitative concentration maps from your own reference spectra.

There are **two** 3-D tools:

- **3D Volume** (§19) — the built-in matplotlib viewer: fast, interactive, good for inspection (scatter / slices / surface / RGB).
- ■ **HQ Volume** (§20–21) — the Plotly publication renderer: smooth solid volumes, multi-component colour, and reference-spectra concentration analysis, with high-resolution PNG export.

Note A real volume exists only when the loaded .wdf has multiple depth (Z) planes. BioRaman detects this automatically (e.g. count = nx·ny·nz) and rebuilds the Z×Y×X volume; the 2-D map view shows the depth-average.

19 3-D Confocal Volume Viewer (matplotlib)

1. Click **3D Volume**.
2. If you have separate per-plane files, click **Load Z-stack WDF files…** and set the **Z step (µm)**; otherwise the current map is used.
3. Choose a **Render mode**: Volume Scatter, Orthogonal Slices, 3D Surface, or Multi-band RGB.
4. Set **Band A/B/C** ranges (cm^{-1}) and the **Threshold**, voxel alpha, Z-scale and pre-smoothing.
5. Click **Render 3D View**; rotate with the mouse. **Export PNG / PDF** at 250 dpi.

Note This viewer is for quick inspection. For a publication figure that looks like a solid filled volume, use the HQ renderer (§20).

20 ◆ Publication Volume Renderer (Plotly)

Opens from the ■ **HQ Volume** toolbar button or **Analysis** → **Publication Volume (Plotly)**. It renders in your web browser (use **Firefox or Chrome** — Safari cannot draw WebGL volumes) and exports a high-resolution PNG.

Render modes

Mode	Look / use
solid block (filled, multi-band)	Filled, opaque coloured block; each voxel painted by its dominant component (the paper-style terrain block). Default.
surface (multi-band, hollow)	Solid iso-surfaces per band, overlaid in colour.
cloud (single band, glow)	Translucent ray-cast of one band.

Step-by-step

1. Click ■ **HQ Volume**.
2. Keep Mode = **solid block (filled, multi-band)**.
3. Set the **Bands** (cm■^{1}) and tick the components you want — each colour is one band (Red / Green / Blue).
4. Set **Z keep** ≥ **/** ≤ **(µm)** to trim empty deep planes to the depth where real signal exists.
5. Tune **Iso level pct** (lower = fills more of the block), **Smoothing** σ ≈1.0–1.4 and **Upsample** = 2–3 for smooth surfaces, and **Z stretch** ≈2–3 for depth.
6. Choose **Background** (white for paper figures).
7. Click ■ **Render (open in browser)**; rotate and inspect.
8. Click ■ **Render & Save PNG…** for the figure (renders headlessly via kaleido — works even if the browser view doesn't).

★ If one side looks empty, that is usually real confocal depth attenuation. Either tick **Equalize depth** to compensate, or set the **Z keep** range to trim the empty planes.

Control	What it does
Iso level pct	Threshold for what is shown / filled (lower = more).
Smoothing σ / Upsample	Smooth and refine the 25^3-ish grid.
Z keep ≥ / ≤	Crop the depth range shown (µm).
Equalize depth	Scale each Z plane to compensate attenuation.
Z stretch	Exaggerate the depth axis.
Surface opacity / Background	Opacity; white or black scene.

21 ✥ Concentration analysis (reference CLS)

This reproduces the Renishaw "concentration estimate" method: every voxel is fit by **non-negative least squares** to your measured reference spectra **without normalisation**, the volume is coloured by the dominant component, poorly-explained voxels are shown as **LoF** (purple), and the overall % of each component is reported.

★ You need your component reference spectra **as data files** (two-column wavenumber,intensity .txt/.csv, or .wdf/.spc) — at least **two** components. With one reference every voxel is trivially 100%.

Step-by-step

1. Load your map/volume with **Preprocessing** → **Normalisation = none** (quantitative fits need raw intensities). Reprocess if needed.
2. Open ■ **HQ Volume**.
3. Click ■ **Load…** and select ALL component reference spectra at once (e.g. PP and LDPE). The label shows e.g. "2: PP, LDPE".
4. Tick **Concentration mode (fit reference spectra)**.
5. Tick **Show lack-of-fit (LoF) in purple** and set the **LoF threshold** (0–1; higher = less purple).
6. Set **Iso level pct** (~45) and the **Z keep** depth window.
7. Click **Render**. The title shows the overall split, e.g. "PP 57.7%, LDPE 42.3%".
8. Use ✕ **Clear** to remove references and start over; ■ **Render & Save PNG** for the figure.

Note The first render performs one NNLS fit per voxel (e.g. 15,625 fits) and takes a few seconds; the result is cached, so changing display settings re-renders quickly. Percentages are relative concentrations (each component's summed abundance ÷ total).

Part IV — Reference

22 Reports, sessions & headless CLI

- **File → Save Analysis Report (HTML)** — self-contained report (map, mean spectrum, recipe, log).
- **File → Save / Load Session** — store source path, recipe and view settings in a .bioses file and restore later.

Headless batch (no GUI):

```
python bioraman.py --input data/ --out processed/ --recipe recipe.json --format .npz
--recursive

python bioraman.py --save-recipe recipe.json
```

23 Exporting results

Output	Where	Format
Map	Save Map (Ctrl+M)	PNG / TXT / NPY
Spectrum	Save Spectrum (Ctrl+S)	TXT / CSV
Processed cube	Save Processed (Ctrl+Shift+S)	NPZ/HDF5/CSV/TXT/MAT
Recipe	Preprocessing → Save Recipe	JSON
QC map	QC Maps window	CSV / NPY
3-D volume	HQ Volume → Save PNG / HTML	PNG / HTML
Report / Session	File menu	HTML / .bioses
Batch	Batch window	per-file + summary CSV

24 Keyboard shortcuts

Shortcut	Action
Ctrl+O	Open a file
Ctrl+M	Save the map
Ctrl+S	Save the spectrum
Ctrl+Shift+S	Save the processed cube
Left / Right click on map	Select pixel / add to comparison

25 Troubleshooting

Map loads as a thin strip (e.g. 15625×1)

renishawWiRE is missing — install it (auto-installs on launch) and reload; the grid (and depth axis) is read from the .wdf header.

3-D render opens but the box is empty

You are probably in Safari — open the saved ..._volume.html in Firefox or Chrome, or use Render & Save PNG (headless).

Concentration shows one component at 100%

Only one reference is loaded. Click ✕ Clear and load at least two component spectra together.

One side of the volume is empty

Real confocal depth attenuation. Tick Equalize depth or set the Z keep range to trim empty planes.

Saving .h5 / .mat fails

Install h5py (HDF5) or SciPy (.mat); or use .npz which needs nothing extra.

26 Glossary

Term	Meaning
Confocal volume	Depth-resolved map: a spectrum at each X·Y·Z point.
Recipe	JSON file of all preprocessing parameters.
CLS / NNLS	Classical least squares / non-negative least squares fit of reference spectra → concentrations.
LoF	Lack of fit — residual not explained by the references (shown purple).
Iso-surface	A surface of constant value through a volume.
Silhouette score	Cluster-separation quality (–1→1).
SNV / vector	Per-spectrum normalisations removing scaling.

Appendix A File-format quick reference

Extension	Type	Backend
.wdf	Renishaw map / volume / spectrum	renishawWiRE
.wip	WITec Project	ramanspy
.txt / .csv / .dpt / .dat / .asc / .jdx	ASCII spectra & Renishaw long format	built-in
.xlsx / .xls	Tabular spectra (PCA)	pandas + openpyxl
.zip	RAMANMETRIX dataset	built-in
Export	NPZ / HDF5 / CSV / TXT / MAT / HTML / PNG	numpy / h5py / scipy / plotly

Appendix B Metadata column reference

Column	Default	Meaning
file	—	File / path pattern the row applies to.
include	True	False drops a spectrum.
standard	False	Calibration-standard spectra.
date / device	—	Acquisition date / instrument.
batch	—	Segments for batch cross-validation.
type	—	Main class/grouping label.